

KWASI GYIMAH-ASANTE

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EXPERIENCE

Fordham University – Department of Computer Science

May 2026 – Present

Undergraduate Research Assistant

New York, NY

- Contributing to a quantum networking research project applying **reinforcement learning** (MaskablePPO) to entanglement scheduling on a quantum switch, learning policies that maximize the rate of high-fidelity end-to-end entangled pairs by choosing when to swap, distill, or wait under decaying-fidelity constraints
- Building the exact optimal-policy baseline via a **value iteration** solver in **Python**, formulating the scheduler as an MDP to benchmark the learned RL policy against the provable optimum across fidelity thresholds, under Prof. Xiaolan Zhang

PROJECTS

HybridRAG | *Python, FastAPI, OpenAI, Anthropic, ChromaDB, BM25, LangChain, Docker*

[GitHub](#)

- Improved retrieval relevance over **BM25**-only search by fusing dense vector search with **BM25** sparse retrieval via **Reciprocal Rank Fusion**, firing both searches concurrently and merging ranked results
- Engineered a **FastAPI** backend implementing a four-stage document pipeline (**ingest** → **index** → **retrieve** → **generate**), using **OpenAI embeddings** and **ChromaDB** for vector storage with idempotent chunk upserts to support safe re-ingestion of updated documents
- Implemented a provider-abstraction layer over the **OpenAI** and **Anthropic** SDKs, enabling citation-grounded answer generation across either model from a unified retrieval pipeline

Codebase Q&A Agent | *Python, Claude API, pgvector, Slack Bolt*

[Live Demo](#) | [GitHub](#)

- Built an **agentic RAG** pipeline where **Claude** autonomously issues up to **6 self-refined tool calls** per query via **tool use**, producing grounded answers with file/line citations
- Engineered an ingestion pipeline that clones **GitHub** repositories, chunks source files into overlapping windows, embeds them with **Voyage AI's** code-specialized embedding model, and stores as 1024-dim vectors in **PostgreSQL** using **pgvector** for cosine similarity search; indexed the **FastAPI** open-source repository (**944 files, 3,241 chunks**) in **31 seconds**
- Built **/ask** and **/ingest Slack Bolt** commands enabling real-time codebase Q&A and on-demand repository indexing, containerized with **Docker** for reproducible deployment

PawPath: AI Dog Walking App | *Python, Gemini, Supabase, Google Maps*

[Live Demo](#) | [GitHub](#)

- Automated health risk alerts for **100+ dog breeds** by integrating **Gemini AI** photo-based breed identification with **Google Maps Platform** (Routes, Places, Weather) for dynamic walk route generation via a **FastAPI** backend
- Built a safety scoring system combining 5 environmental signals (temperature, AQI, UV index, humidity, wind) to assess walk conditions and surface breed-specific health risk alerts
- Built a **React** frontend with live GPS tracking, dog profile management, and a breed-specific exercise recommendation engine, backed by full **CRUD** on **Supabase**

EDUCATION

Fordham University

August 2024 – May 2028

B.Sc. in Mathematics & Computer Science

New York, NY

- Relevant Coursework:** Computer Algorithms, Data Structures, Discrete Mathematics, Calculus I, II, III

CompTIA Security+

August 2025

Security Concepts, Threats & Mitigations, Security Architecture, Security Operations

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL (PostgreSQL)

AI & ML: LLM Integration, RAG Pipelines, AI Agent Design & Tool Use, Embeddings & Vector Databases, Prompt & Context Engineering, Model Evaluation, Reinforcement Learning, RLHF, Fine-Tuning (LoRA/QLoRA)

Frameworks & Libraries: Flask, HuggingFace, LangChain, OpenAI/Anthropic SDKs, Pydantic, NumPy

Backend & Infrastructure: FastAPI, Node.js, PostgreSQL, Docker, OAuth 2.0

Tools & Platforms: Git, GitHub Actions, Railway, AWS, Vercel, Supabase, Linux